

Handout 1: Initial Literature Review

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1 Student Outcomes

By the end of this activity, students will be able to:

1. Search for and identify relevant research papers using tools such as Google Scholar and LLM-assisted search methods.
2. Distinguish between different types of research publications including review papers, journal papers, conference papers, and arXiv preprints.
3. Efficiently perform an initial literature review by skimming abstracts, figures, and introductions to identify relevant prior work and collect key references.

2 Equipment

1. Use Google Scholar to search for research papers related to your project topic.
<https://scholar.google.com>
2. Accessing research papers.
 - (a) Students working on campus and connected to UIC WiFi can usually download papers directly through the university library subscriptions.
 - (b) Students working off campus can access papers using one of the following methods:
 - i. Use the UIC VPN service to connect to the university network. Instructions are available here:
<https://it.uic.edu/services/faculty-staff/uic-network/uic-vpn/>
 - ii. Use the UIC Library EZProxy service. Instructions are available here:
<https://library.uic.edu/help-articles/proxy-urls-ezproxy-2/>
3. Gemini is highly recommended among all LLMs.

3 Initial Literature Review

1. Search for papers using keywords on Google Scholar.

<https://scholar.google.com>

Alternatively, you can use Large Language Models (LLMs) such as Gemini or ChatGPT to help generate search queries and identify papers. However, be aware that LLMs may sometimes generate fake or incorrect references. Always verify papers by searching for them on Google Scholar.

2. Research papers generally fall into several categories:

- (a) **Review Papers:** Papers that summarize and review the state of a research field. These are useful for obtaining a broad overview of a topic.
- (b) **Journal Papers:** Typically more complete and mature research works. These papers often contain detailed experiments, analysis, and sometimes supplementary videos or supplementary documents.
- (c) **Conference Papers:** Usually shorter papers that may present preliminary or complete research results. Many robotics and engineering results first appear in conferences.
- (d) **arXiv Papers:** Preprints uploaded by authors before formal peer review or publication. These can provide access to the latest research developments.

3. How to Read Papers

For the initial literature review, focus on breadth rather than depth. The goal is to quickly identify relevant papers and understand the landscape of the field.

- (a) Read the abstract carefully.
 - (b) Skim through the figures, tables, and captions to understand the main ideas and results.
 - (c) Check whether the paper is relevant to your project.
 - (d) Skim through the introduction section to identify additional references and related work.
4. At this stage, your goal is to collect and organize approximately 20 relevant papers. Save the PDFs and maintain a list of references for future reading. You can organize the papers based on topics in separate folders or maintain a text file.

Show your collection of relevant literature to the teaching assistant.